TEK4090

Introduction to Modern Control

Lecture 3: State Space Systems

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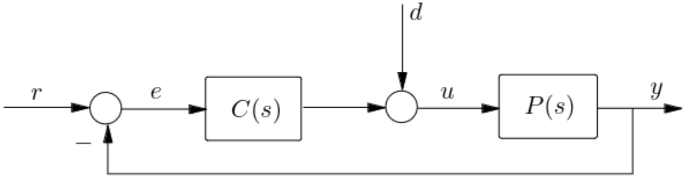
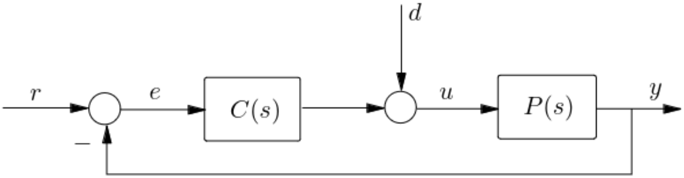
Reading Assignment

1. Francis: Ch 4.5, Ch 5
2. Åstrom & Murray: Ch 5-6, Ch 7.1

Standard Feedback Control



Theorem of Stability for Feedback Control



Theorem

The closed-loop feedback system is stable if and only if the zeros of the **characteristic polynomial**

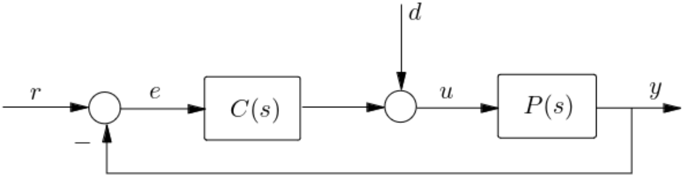
$$D_p D_c + N_p N_c \tag{1}$$

are all in the open left-hand plane.

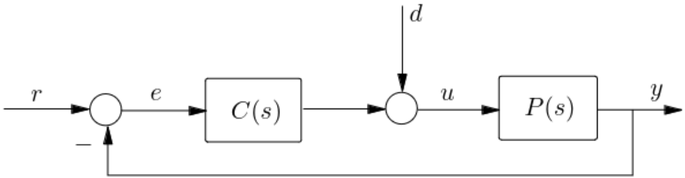
Example 1

Example 2

We saw the transfer function from e to u...



All the Transfer Functions



$$\begin{bmatrix} E \\ U \end{bmatrix} = \begin{bmatrix} \frac{1}{1+PC} & -\frac{P}{1+PC} \\ \frac{C}{1+PC} & \frac{1}{1+PC} \end{bmatrix} \begin{bmatrix} R \\ D \end{bmatrix} \tag{2}$$

Eg., transfer function from D to E is:

$$-\frac{P}{1+PC} \tag{3}$$

Reference Tracking: Motivation

Reference Tracking: Final Value

Theorem

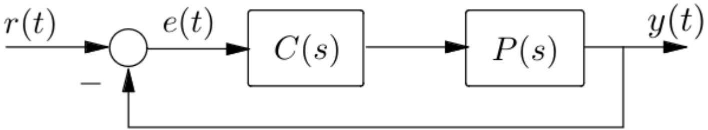
Consider a signal $f(t)$ with (rational, strictly proper) Laplace transform $F(s)$

- If $F(s)$ has no poles in $\Re(s) \geq 0$, then $f(t)$ has a final value: $\lim_{t\to\infty} f(t) = 0$
- If $F(s)$ has no poles in $\Re(s) \geq 0$ except a simple pole at $s = 0$, then

$$\lim_{s\to 0} sF(s) \tag{4}$$

- In all other cases, f is unstable, so $\lim_{t\to\infty} f(t) = \infty$

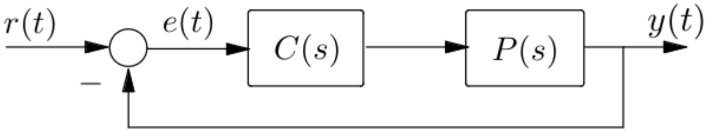
Reference Tracking



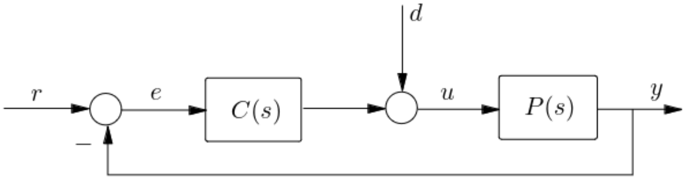
$$P(s) = \frac{1}{s+1}$$
$$C(s) = 1$$
$$R(s) = \frac{100}{s}$$

This is a simple model for cruise control

Reference Tracking

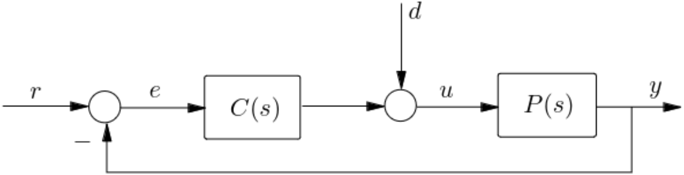


Recap from Last Time



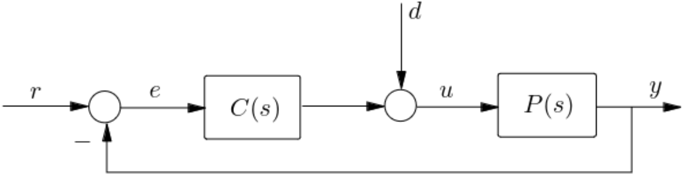
- Plant $P(s)$: Laplace-domain model of our system
- Controller $C(s)$: Takes measurement $y(y)$ and modifies input $u(t)$
- Reference $r(t)$: A pre-determined signal that tells the system what to do

Recap: Group of Four



$$\begin{bmatrix} E \\ U \end{bmatrix} = \begin{bmatrix} \frac{1}{1+PC} & -\frac{P}{1+PC} \\ \frac{C}{1+PC} & \frac{1}{1+PC} \end{bmatrix} \begin{bmatrix} R \\ D \end{bmatrix} \tag{5}$$

Recap: Stability



Stability was equivalent to:

$$N_p N_c + D_p D_c$$

having no zeros in RHP.

$$\begin{bmatrix} E \\ U \end{bmatrix} = \begin{bmatrix} \frac{1}{1+PC} & -\frac{P}{1+PC} \\ \frac{C}{1+PC} & \frac{1}{1+PC} \end{bmatrix} \begin{bmatrix} R \\ D \end{bmatrix}$$

Today

- Nyquist Stability Criterion
- Nyquist Plots
- Bode Plots
- State-Space Systems
- Reachability/Observability

Recall Complex Integration

The integral of f in a closed curve in $\mathbb{C}$ is related to the **singularities** of f.

$$\oint_{\mathcal{C}} f(s)ds = 2\pi i \sum_{z_k \in \mathcal{C}} \mathrm{res}(f, z_k) \tag{6}$$

The residues of poles are easily computed.

Stability

The main idea:

- Stability of the feedback system is determined by RHP poles of

$$\frac{1}{1 + P(s)C(s)}. \tag{7}$$

- This is equivalent to zeros of the characteristic polynomial

$$N_p N_c + D_p D_c \tag{8}$$

- We want a nice graphical test of whether or not a system is stable
- **Idea:** can we use complex integration to tell us something?

The Argument of a Complex Number

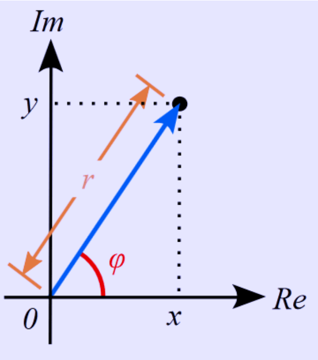
Polar representation of complex number:

$$z = a + ib = re^{i\phi}. \tag{9}$$

The angle ϕ is also called the **argument**.

Taking the log:

$$i\phi = \ln \frac{z}{r} \tag{10}$$



How does the argument of a function change?

$$i \arg z = \ln \frac{z}{|z|}$$

Recall:

$$\frac{d}{dz} \ln f(z) = \frac{f'(z)}{f(z)}$$

This gives us the 'rate of change' of the argument

The total change of argument on a curve can be computed as,

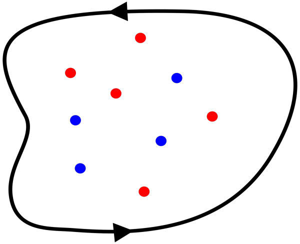
$$\frac{1}{i} \oint_{\mathcal{C}} \frac{f'(z)}{f(z)} dz \tag{11}$$

For 'nice' functions, this counts zeros and poles!

Cauchy's Principle of the Argument

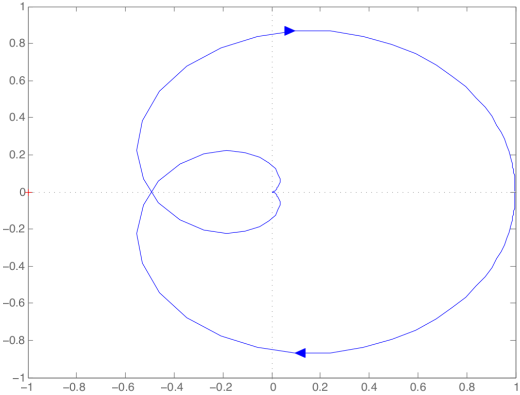
Let G be a meromorphic function, and $\mathcal{C}$ be a closed CW curve in $\mathbb{C}$. Then,

$$\begin{aligned}\Delta \arg G(s) &= \frac{1}{2\pi i}\oint_{\mathcal{C}} \frac{G'(z)}{G(z)} dz \\ &= \sum_{p_k \in \mathcal{C}} (1) - \sum_{z_k \in \mathcal{C}} (1) \\ &= n - m\end{aligned}$$



What is this saying?

How **windy** a function is on $\mathcal{C}$ tells us how many poles and zeros it has in the RHP

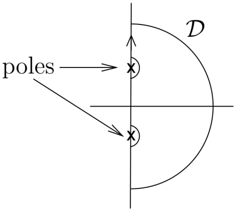


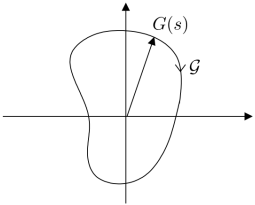
This function **winds** around 0 'negative' twice CCW, so $n - m = 2$

If we additionally know that this function has two zeros in $\mathcal{C}$, then we know that it has zero poles inside $\mathcal{C}$.

Nyquist Stability Criterion

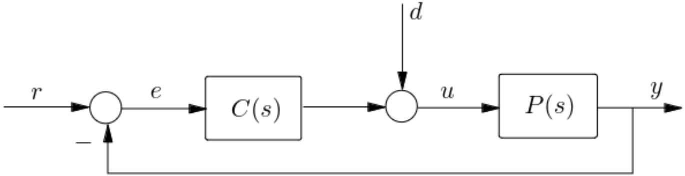
We care about poles/zeros in the RHP for stability





- Take a closed curve $\mathcal{D}$ in RHP
- Look at its image $\mathcal{G}$ under $G(s)$
- **Argument principle:** How many times $G(s)$ circles around the origin is (#poles - #zeros) in $\mathcal{D}$.

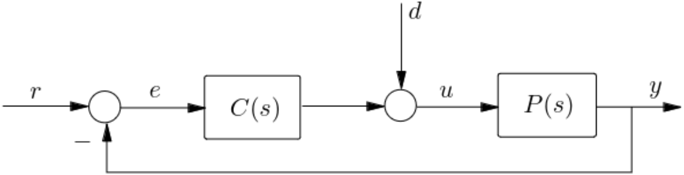
Setup



Assumptions:

1. $P(s), C(s)$ are proper with at least one strictly proper
2. $P(s)C(s)$ has no pole-zero cancellations in $\Re(s) \geq 0$

Nyquist Criterion



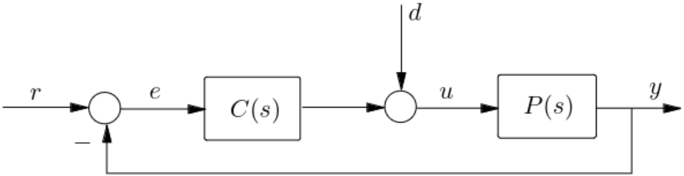
$$\begin{bmatrix} E \\ U \end{bmatrix} = \begin{bmatrix} \frac{1}{1+PC} & -\frac{P}{1+PC} \\ \frac{C}{1+PC} & \frac{1}{1+PC} \end{bmatrix} \begin{bmatrix} R \\ D \end{bmatrix}$$

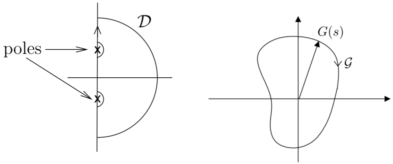
- No pole-zero cancellations:

  $$D_p D_c + N_p N_c, \ 1 + P(s)C(s) := G(s)$$

  have the same RHP zeros
- Stability $\iff$ $G(s)$ no RHP zeros

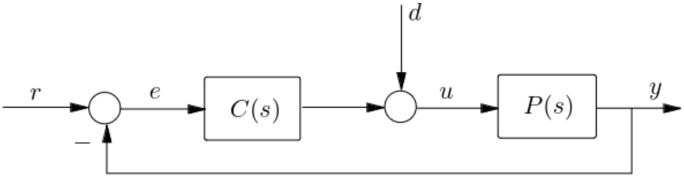
Nyquist Criterion





- Proper: $G(s)$ has more poles than zeros
- n poles, m zeros in RHP
- Stability $\iff m = 0$
- $G(s)$ circles origin n times CCW
- Alternately, $P(s)G(s)$ circles -1, n times CCW

Nyquist Criterion: Final Statement



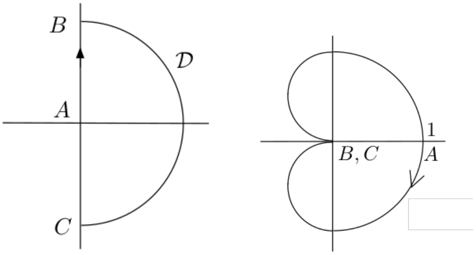
$$\begin{bmatrix} E \\ U \end{bmatrix} = \begin{bmatrix} \frac{1}{1+PC} & -\frac{P}{1+PC} \\ \frac{C}{1+PC} & \frac{1}{1+PC} \end{bmatrix} \begin{bmatrix} R \\ D \end{bmatrix}$$

- Let n denote the number of poles of $P(s)C(s)$ in $\Re(s) \geq 0$
- The feedback system is stable iff the Nyquist plot doesn't pass through -1 and encircles it exactly n times CCW

Nyquist Logic Summarized

- The feedback loop is stable if the characteristic polynomial
$$D_p D_c + N_p N_c \tag{12}$$
has no RHP zeros.
- Equivalent to $G(s) = 1 + P(s)C(s)$ having no zeros
- Principle of the argument says that $\mathcal{G} = 1 + P(s)C(s)$ encircles the origin (or, $P(s)C(s)$ encircles -1) $n - m$ times CCW
- Therefore, $m = 0$ ($G(s)$ has no RHP zeros) iff $\mathcal{G}$ encircles the origin n times CCW
- Therefore, feedback stability iff $P(s)C(s)$ encircles -1 exactly n times CCW

Nyquist Plot Example 1



$$P(s)C(s) = \frac{1}{(s+1)^2}$$

On segment $A \to B$, $s = j\omega$, and so

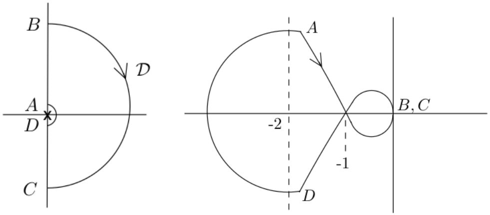
$$\Re PC = \frac{1-\omega^2}{(1-\omega^2)^2 + (2\omega)^2}$$

$$\Im PC = \frac{-2\omega}{(1-\omega^2)^2 + (2\omega)^2}$$

$PC(0) = 1.$

As s approaches B, $PC(j\omega) \mapsto -1/\omega^2$

Nyquist Plot Example 2



$$P(s)C(s) = \frac{s+1}{s(s-1)}$$

(Just use Matlab...)

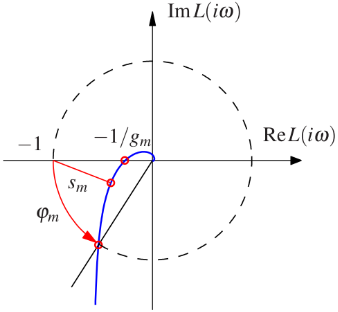
Nyquist Plots

Nyquist Properties

- Scales by constants:
  - Eg., constant gain $C(s) = kH(s)$
  - If the plot crosses -1 for some K, then we can see this easily
- Otherwise, not super useful for **designing** controllers

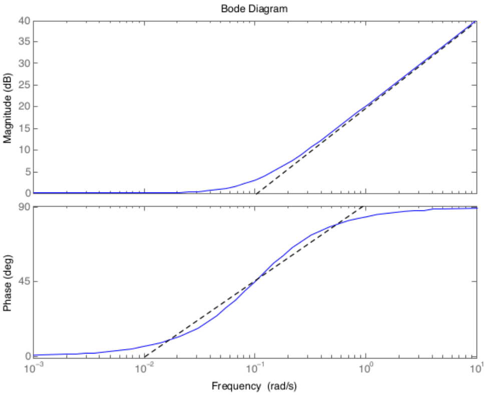
Nyquist Plots

Punchline: Encirclements (or de-encirclements) stabilize (or destabilize) the system



1. **Gain Margin:** g_m: how much we can scale the plot
2. **Phase Margin:** how much phase a $C(s)$ can add
3. **Stability Margin:** shortest distance to -1

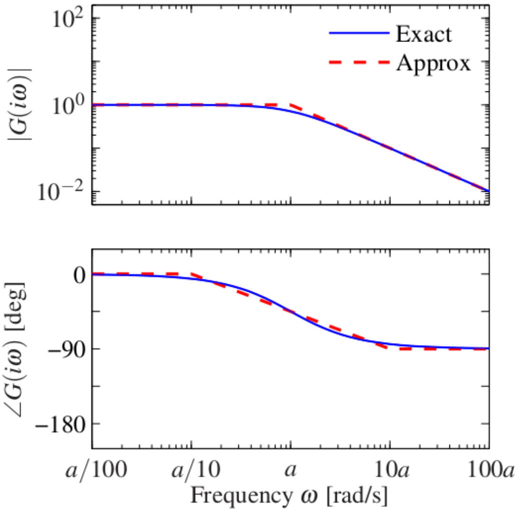
Bode Plots



$$G(s) = 10s + 1 \tag{13}$$

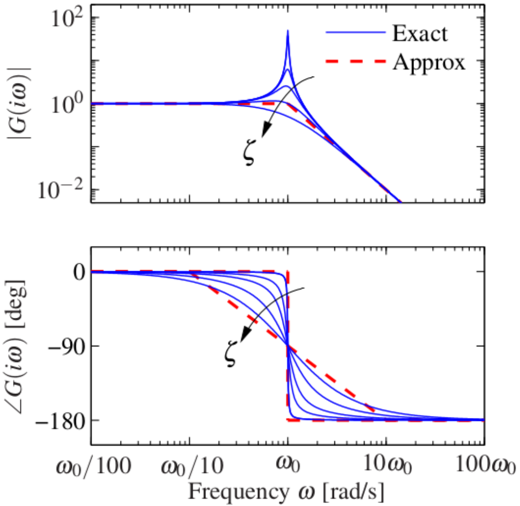
1. **Magnitude** (in dB): $|G(j\omega)|$
2. **Phase** (in deg): $\arg G(j\omega)$

Bode Plots: Examples



$$G(s) = \frac{a}{s+a} \tag{14}$$

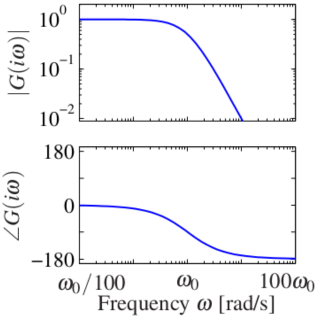
Bode Plots: Examples

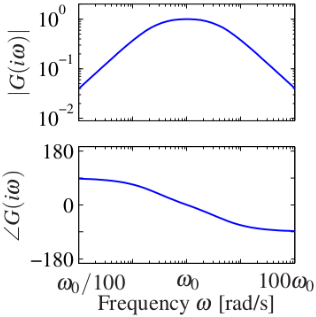


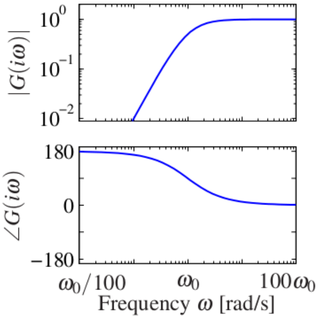
$$G(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2} \tag{15}$$

- Resonant peak at ω_0
- Height of peak: damping ζ

Bode Plots: Filters





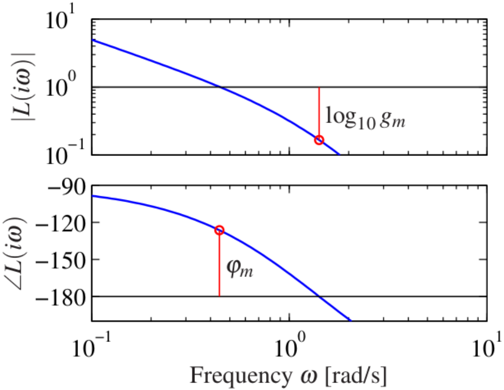


$$G(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

$$G(s) = \frac{2\zeta\omega_0 s}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

$$G(s) = \frac{s^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

Stability margins



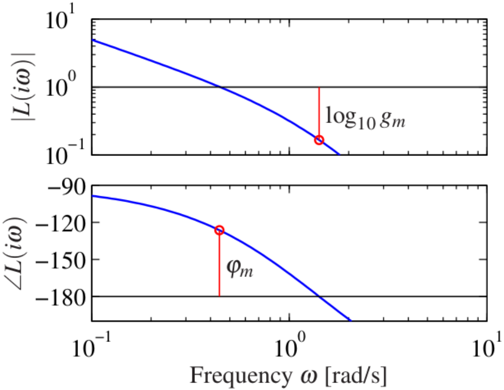
- **Gain Crossover Frequency:**

$$\omega_{gc} := \min_{\omega \in \mathbb{R}} \{ |L(i\omega)| = 10\text{dB} \}$$

- **Phase Margin:**

$$\varphi_m := \pi + \arg L(i\omega_{gc})$$

Stability margins



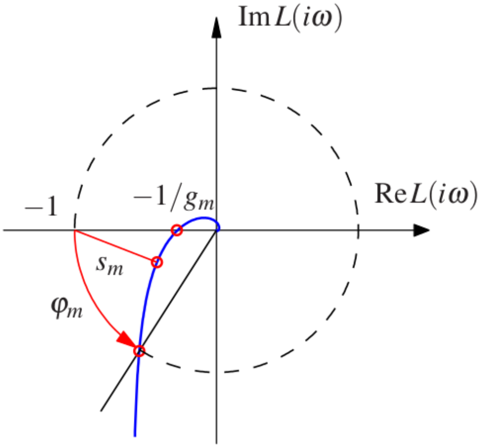
- **Phase Crossover Frequency:**

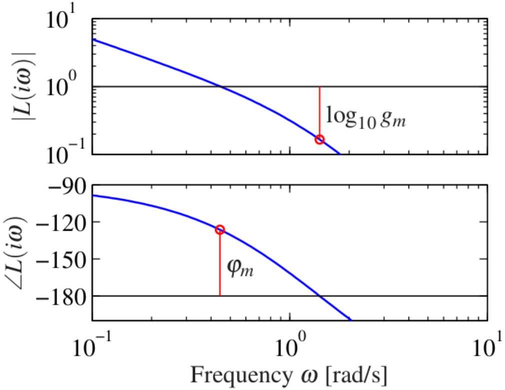
$$\omega_{pc} := \min_{\omega \in \mathbb{R}} \{\arg L(i\omega) = 180^\circ\}$$

- **Gain Margin:**

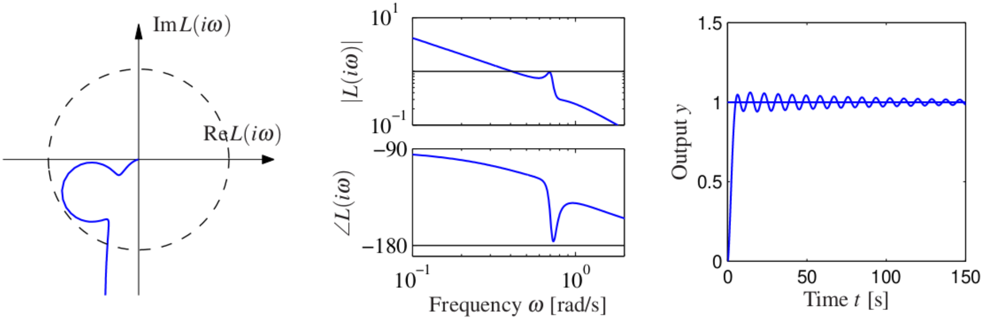
$$g_m := \frac{1}{|L(i\omega_{pc})|}$$

Nyquist vs Bode





Good Gain/Phase Margin, Poor Stability Margin



State-Space Systems

Linear Systems

$$\dot{x} = Ax + Bu \tag{16}$$

$$y = Cx \tag{17}$$

Matrix Exponential

Defined in terms of its Taylor expansion

$$e^a = 1 + a + \frac{a^2}{2!} + \frac{a^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{a^k}{k!} \tag{18}$$

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k}{k!} \tag{19}$$

So, the derivative:

$$\frac{d}{dt} e^{At} = \frac{d}{dt} \left(I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \cdots \right) = A e^{At} \tag{20}$$

Existence of Solutions

Theorem

A solution to the ODE is given by

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \tag{21}$$

Proof: Take derivative w.r.t t

Diagonalizable Matrices

Theorem

A matrix A is called **diagonalizable** if there exists T such that

$$D = TAT^{-1}, \tag{22}$$

where D is a diagonal matrix containing the eigenvalues of A.

Then, we can write the system matrix exponential as,

$$e^{At} = Te^{Dt}T^{-1}. \tag{23}$$

Non-Diagonalizable Matrices

Complicated, but happens when you have repeated eigenvalues

$$A = TJT^{-1}$$

Jordan form:

$$J = \begin{bmatrix} J_1 & 0 & \cdots & 0 & 0 \\ 0 & J_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & J_{k-1} & 0 \\ 0 & 0 & \cdots & 0 & J_k \end{bmatrix} \qquad J_i = \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda_i & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_i & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda_i \end{bmatrix}$$

Matrix Exponential of Jordan Form

$$e^{A} = Te^{J}T^{-1} \tag{24}$$

$$e^{J} = \begin{bmatrix} e^{J_1} & & & \\ & e^{J_2} & & \\ & & \ddots & \\ & & & e^{J_k} \end{bmatrix} \qquad e^{J_i} = e^{\lambda_i} \begin{bmatrix} 1 & t & \frac{t^2}{2} & \cdots & \frac{t^{n-1}}{(n-1)!} \\ 0 & 1 & t & \cdots & \frac{t^{n-2}}{(n-2)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & t \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Eigenvalues and Modes

Coordinate Invariance

The coordinate frame chosen to represent the states/inputs x and u affects the values of A, B and C.

Recall from the linear algebra class that a map $z = Tx$ with T invertible can represent a linear coordinate shift.

$$\frac{dz}{dt} = T(Ax + Bu) = TAT^{-1}z + TBu \tag{25}$$

$$y = Cx + Du = CT^{-1}z + Du \tag{26}$$

or

$$\dot{z} = \tilde{A}z + \tilde{B}u, \;\; \tilde{A} = TAT^{-1}, \;\; \tilde{B} = TB \tag{27}$$

$$y = \tilde{C}z + Du, \;\; \tilde{C} = CT^{-1} \tag{28}$$

Coordinate Invariance

Using the property

$$e^{TST^{-1}} = Te^{S}T^{-1}, \tag{29}$$

it can be shown that

$$x(t) = T^{-1}z(t) = T^{-1}e^{\tilde{A}t}Tx(t) + T^{-1}\int_0^t e^{\tilde{A}(t-\tau)}\tilde{B}u(\tau)d\tau \tag{30}$$

Steady-State Response

Output system:

$$y(t) = Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}Bu(\tau)d\tau \tag{31}$$

Nonlinear systems & Linearization

$$\frac{dx}{dt} = f(x,u),\ x \in \mathbb{R}^n, u \in \mathbb{R}$$
$$y = h(x,u),\ y \in \mathbb{R}$$

Consider an **equilibrium point** (x_e, u_e) such that $f(x_e, u_e) = 0$, and variables such that

$$z = x - x_e$$
$$v = u - u_e$$
$$w = y - h(x_e, u_e)$$

The **Jacobian linearization** of the system is,

$$\frac{dz}{dt} = Az + Bv,\ w = Cz + Dv$$
$$A = \left.\frac{\partial f}{\partial x}\right|_{(x_e,u_e)},\ B = \frac{\partial f}{\partial u}\Big|_{(x_e,u_e)}$$
$$C = \frac{\partial h}{\partial x}\Big|_{(x_e,u_e)},\ D = \frac{\partial h}{\partial u}\Big|_{(x_e,u_e)}$$

Example: Vehicle Steering

Example: Vehicle Steering

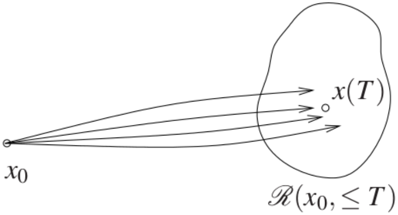
Reachability

Quick Note

For clarity, we will consider systems with a single input and single output (SISO), i.e. $u(t), y(t) \in \mathbb{R}$, and so $B \in \mathbb{R}^{n\times 1}$ and $C \in \mathbb{R}^{1\times n}$.

All the final results hold if you consider MIMO systems as well

Reachability



The **reachable set** $\mathcal{R}(x_0, T)$ is the set of all points x_f such that there exists an input $u(t)$ that steers the system from $x(0) = x_0$ to $x(T) = x_f$.

A system is **reachable** if for any $x_0, x_f \in \mathbb{R}^n$, there exists $T > 0$ and $u : [0, T] \mapsto \mathbb{R}$ such that $x(0) = x_0$ and $x(T) = x_f$.

Test for Reachability

Test for Reachability

Test for Reachability

Test for Reachability

Theorem

A linear system is reachable if and only if the reachability matrix W_r is invertible.

Reachable Canonical Form

$$\frac{dz}{dt} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & & & 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u,$$

$$y = \begin{bmatrix} b_1 & b_2 & b_2 & \cdots & b_n \end{bmatrix} z + Du$$

$$W_r = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}$$
$$= \begin{bmatrix} 1 & -a_1 & a_1^2 - a_2 & \cdot & * \\ 0 & 1 & -a_1 & \cdot & * \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

How do we convert a system to this form?

Conversion to Reachable Canonical Form

Conversion to Reachable Canonical Form

Conversion to Reachable Canonical Form

Summary of Today

- Nyquist Stability
- Stability Margins
- Nyquist & Bode Plots
- Linear State-Space Systems Theory

Next Time:

- MATLAB Lecture
- Some tools for transfer functions and Bode/Nyquist plots